

DIGITAL PAYMENT TRANSACTION ANALYTICS

1. Problem Statement

PayFlow is a fictional digital payments company that processes transactions for customers and merchants. The company wants a centralized relational database to manage customers, merchants, payment methods, transactions, refunds, settlements, transaction fees and transaction-risk information. Analysts must use SQL to monitor payment performance, customer and merchant behavior, failures, refunds, settlements and potentially risky activity.

The assignment preserves the technical focus of the uploaded Employee Management System case study database design, normalization, constraints, SQL queries, triggers, transactions, string manipulation and date/time functions while changing the business domain to digital payments.

2. Objectives

- Design the database in at least 3 Normal Form.
- Use Primary Key / Foreign Key, NOT NULL, UNIQUE, CHECK and DEFAULT constraints.
- Load and validate realistic payment data.
- Answer business questions using SQL.
- Demonstrate joins, aggregation, CASE, subqueries, CTEs and window functions.
- Analyze transaction success/failure, customer behavior, merchant performance, refunds, settlements and risk.
- Translate SQL outputs into business recommendations without relying on a BI tool.

3. Tables and Exact Schema

Customer

CustomerID PK; FirstName NOT NULL; LastName NOT NULL; Email UNIQUE NOT NULL; PhoneNumber UNIQUE NOT NULL; RegistrationDate NOT NULL; City; CustomerStatus.

Merchant

MerchantID PK; MerchantName UNIQUE NOT NULL; BusinessType NOT NULL; City; RegistrationDate NOT NULL; MerchantStatus.

PaymentMethod

PaymentMethodID PK; CustomerID FK; MethodType; Provider; AccountIdentifier; IsActive.

Transaction

TransactionID PK; CustomerID FK; MerchantID FK; PaymentMethodID FK; TransactionDate; Amount > 0; TransactionType; TransactionStatus; PaymentChannel; TransactionReference UNIQUE; FailureReason.

Refund

RefundID PK; TransactionID FK; RefundAmount > 0; RefundDate; RefundStatus; RefundReason.

Settlement

SettlementID PK; MerchantID FK; TransactionID FK UNIQUE; SettlementAmount > 0; SettlementDate; SettlementStatus; SettlementReference UNIQUE.

TransactionFee

FeeID PK; TransactionID FK; FeeType; FeeAmount >= 0; FeeDate.

TransactionRisk

RiskID PK; TransactionID FK UNIQUE; RiskScore 0–100; RiskLevel; ReviewStatus; ReviewDate.

4. Data Requirements

The SQL dataset contains 256 rows in total, including 80 transaction records. It is designed for MySQL 8.0+ and contains realistic relationships and deliberate operational patterns for analysis.

- Customers: 15
- Merchants: 10
- Payment methods: 22
- Transactions: 80
- Refunds: 17
- Settlements: 13
- Transaction fees: 19
- Risk records: 80

The original source also specifies referential integrity, normalization to at least 3 Normal Form, and constraints such as NOT NULL, UNIQUE, CHECK and DEFAULT; those principles are retained here.

5. SQL Questions — Basic

1. Display all customers from Mumbai.
2. Display all active merchants.
3. List all successful transactions.
4. Find transactions with Amount > ₹2,000.
5. Display transactions made through UPI.
6. Display failed transactions with their failure reasons.
7. Find customers registered after 1 March 2025.
8. Find minimum, maximum and average transaction amount.
9. Count total transactions.

10. Count transactions by status.
11. Calculate total transaction value.
12. Calculate transaction count and total value by payment channel.

6. SQL Questions — Intermediate

13. Join Transaction and Customer to show customer name, date, amount and status.
14. Join Transaction and Merchant to show merchant name and transaction value.
15. Find the top 5 merchants by successful transaction value.
16. Find transaction count and total value by merchant.
17. Find average transaction value by merchant.
18. Calculate success rate by payment method type.
19. Calculate failure rate by payment channel.
20. Find the most common failure reason.
21. Find customers with more than 5 transactions.
22. Find customers whose total transaction value exceeds ₹10,000.
23. Find merchants with more than 3 failed transactions.
24. Find monthly transaction volume and transaction value.
25. Find daily transaction volume.
26. Calculate completed refund amount by merchant.
27. Find merchants with the highest refund count.
28. Calculate settlement amount by settlement status.
29. Find pending settlement amount by merchant.
30. Calculate total processing fee collected.

7. SQL Questions — Advanced

30. Calculate total processing fee collected.
31. Find the top 5 customers by successful transaction value.
32. Find the average transaction amount for each customer.
33. Find customers who have both successful and failed transactions.

34. Find the merchant with the highest average successful transaction value.
35. Calculate the success rate for each merchant.
36. Find merchants where failed transactions are more than successful transactions.
37. Find the payment channel generating the highest successful transaction value.
38. Compare average transaction value across payment channels.
39. Find the most common failure reason for each payment channel.
40. Find customers whose total successful transaction value is above the overall average customer transaction value.

Refund & Settlement Analysis

41. Find the total refunded amount and refund count for each merchant.
42. Find merchants where refunded amount is greater than the average refund amount across all merchants.
43. Calculate the refund percentage for each merchant based on successful transaction value.
44. Find merchants with pending settlements and their pending settlement amount.
45. Compare successful transaction value with settled amount for each merchant.

Risk Analysis

46. Find the number of high-risk transactions for each merchant.
47. Compare transaction status across different risk levels.

Fee Analysis

48. Find the merchants generating the highest processing fee.
49. Calculate the average processing fee per successful transaction for each merchant.
50. Find the top 5 merchants based on successful transaction value, along with transaction count, refund amount and processing fees.

8. Expected Business-Analysis Outputs

Transaction Health: Aggregations on overall transaction volume, gross merchandise value (GMV), overall success/failure counts, and metrics evaluating transaction amounts.

Payment Method Performance: Groupings across payment channels and payment method types analyzing volume, total value, average ticket size, and channel-specific success versus failure percentages.

Merchant Performance: Joins between merchant and transaction records calculating total value, average transaction size, high-volume merchants, failure ratios per merchant, and composite summaries including refunds and fees.

Customer Behavior: Customer spending summaries identifying high-value users, individual purchase frequencies, average spend per customer, preference by payment method, and customers experiencing mixed success/failure statuses.

Failure Analysis: Failure logs filtering failed transactions, ranking primary failure reasons platform-wide, and isolating top failure reasons broken down by specific payment channel.

Refund Analysis: Refund table joins measuring total completed refund values per merchant, refund counts, merchants exceeding average refund thresholds, and calculating refund percentages relative to successful GMV.

Settlement Monitoring: Settlement status aggregations comparing total settled, pending, and failed amounts, pinpointing merchants with pending settlements, and auditing successful capture amounts against settled balances.

Risk Monitoring: Transaction risk joins evaluating high-risk transaction counts and exposed value per merchant, alongside cross-analyses comparing transaction completion status across risk score levels.

Trend Analysis: Date/time truncations and groupings calculating monthly transaction volume and value trends alongside daily transaction distribution patterns.

Operational Exceptions: Processing fee aggregations calculating total fees collected, top fee-generating merchants, average fee per successful payment, and cross-table audits identifying successful-but-unsettled transactions and velocity anomalies.